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Printed pages: 02 **Sub Code: NEC-024R**

Paper Id: **131634** **Roll No:** | | | | | | | | | | |

B.TECH
(SEM VI) THEORY EXAMINATION 2017-18
ADVANCED DIGITAL SYSTEM DESIGN USING VERILOG

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections.
2. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x10 = 20

a. What do you understand by identifiers and memory operators?
b. Draw a simulation flow chart.
c. What do you understand by assignments with delays?
d. What is sequential circuit testing?
e. What are the differences between behavioral modeling and switch level modeling?
f. Define initial construct.
g. Distinguish between a task and a function.
h. What are test bench techniques?
i. What is fault detection?
j. Define bidirectional gates.

SECTION B

2. Attempt any *three* of the following: 10 x 3 = 30

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(a) With the help of suitable examples explain concurrent and procedural statements with syntaxes. Also with the help of block diagram explain the components of a verilog module.

(b) What is a tri- state gate and explain each type of tri state gate with truth tables.Also design module and a test bench for a half adder?

(c) Discuss the differences between blocking and non blocking assignments with suitable exdamples. Design a half adder using CMOS switches. What are basic switch primitives?

(d) What do you understand by user defined primitives (UDP) and explain their types with suitable examples.

(e) With the help of suitable examples explain the concept of formal verification of a system. Also explain the use of assert cycle sequence and assert next with suitable examples.

SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

(a) What is the need for synthesis? Briefly discuss the two common approaches for hardware realization using synth ite the structure of typical simulation module and explain.

(b) Explain different data types i lso discuss the different driving strengths in verilog with strengt and element modeled.

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4. Attempt any *one* part of the following: 10 x 1 = 10

(a) Explain 'wand' and 'wor' types of nets with examples. Also explain NAND gate primitive with verilog module.

(b) Draw a 3 to 8 decoder using 2 to 4 decoder and write the verilog module for the same.

5. Attempt any *one* part of the following: 10 x 1 = 10

(a) Design a module for 4 bit four functions ALU and also a test bench for the same by using behavioral modeling.

(b) Design a 2:4 demultiplexer module and test bench using the if-else-if construct in a verilog.

6. Attempt any *one* part of the following: 10 x 1 = 10

(a) Explain why 'specparam' construct is used in specify block? Distinguish between 'specparam' and 'parameter'. Also explain recursive function with example.

(b) Briefly explain feedback model and a capacitive model with suitable examples. Also write verilog module for D latch using UDP.

7. Attempt any *one* part of the following: 10 x 1 = 10

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(a) Draw the basic functional unit of a dynamic shift register using switch level modeling. Also write verilog module and test bench.

(b) With the help of suitable examples briefly explain BDD, ordered BDD and LPDD. Also write the differences between them.

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